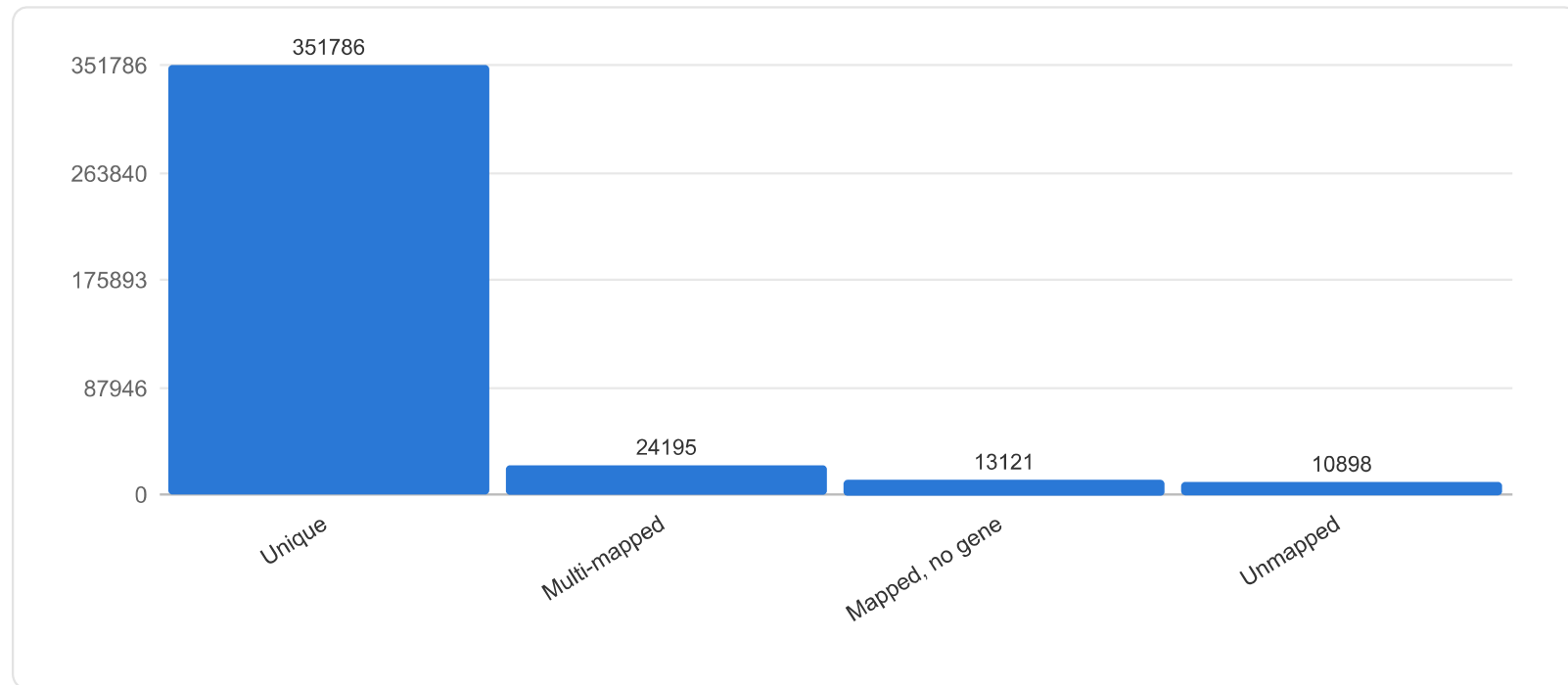


rnaseq_pipeline.c — run report

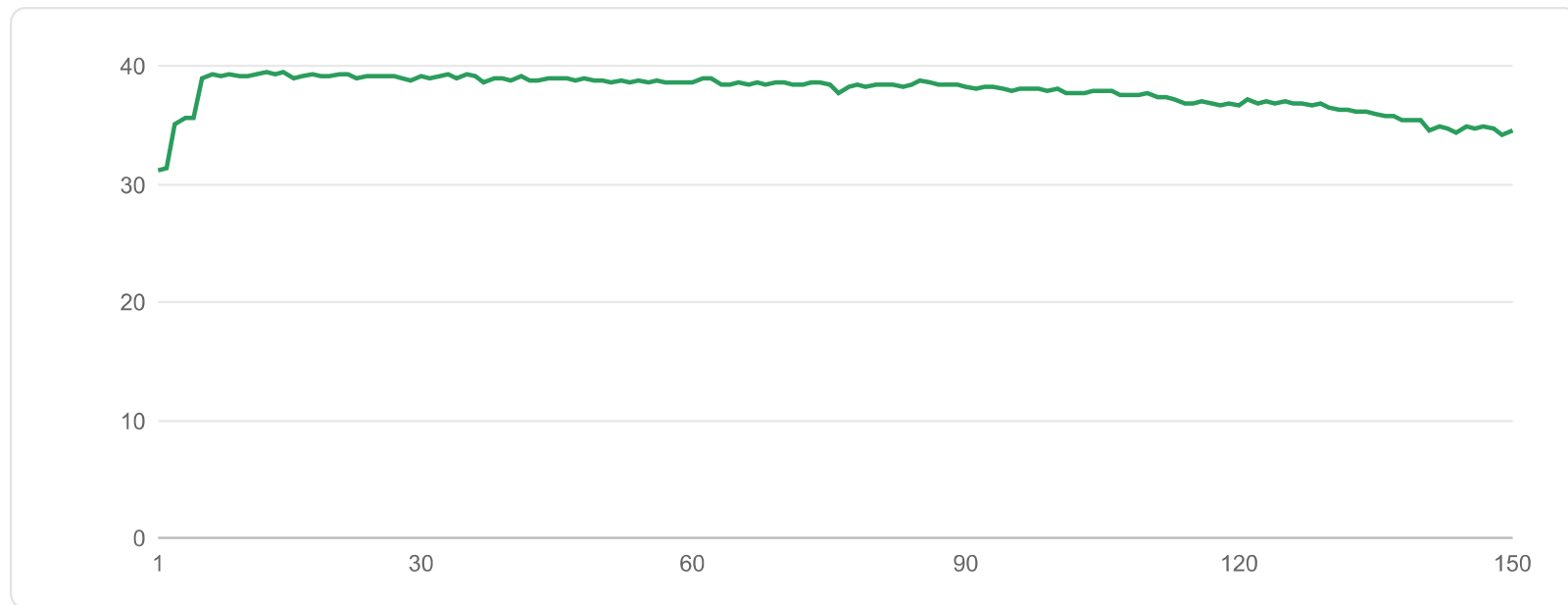
sub400k/r1.fastq + sub400k/r2.fastq · 400000 read pairs · single-sample run (see note at bottom on sample-comparison plots)

Alignment fate



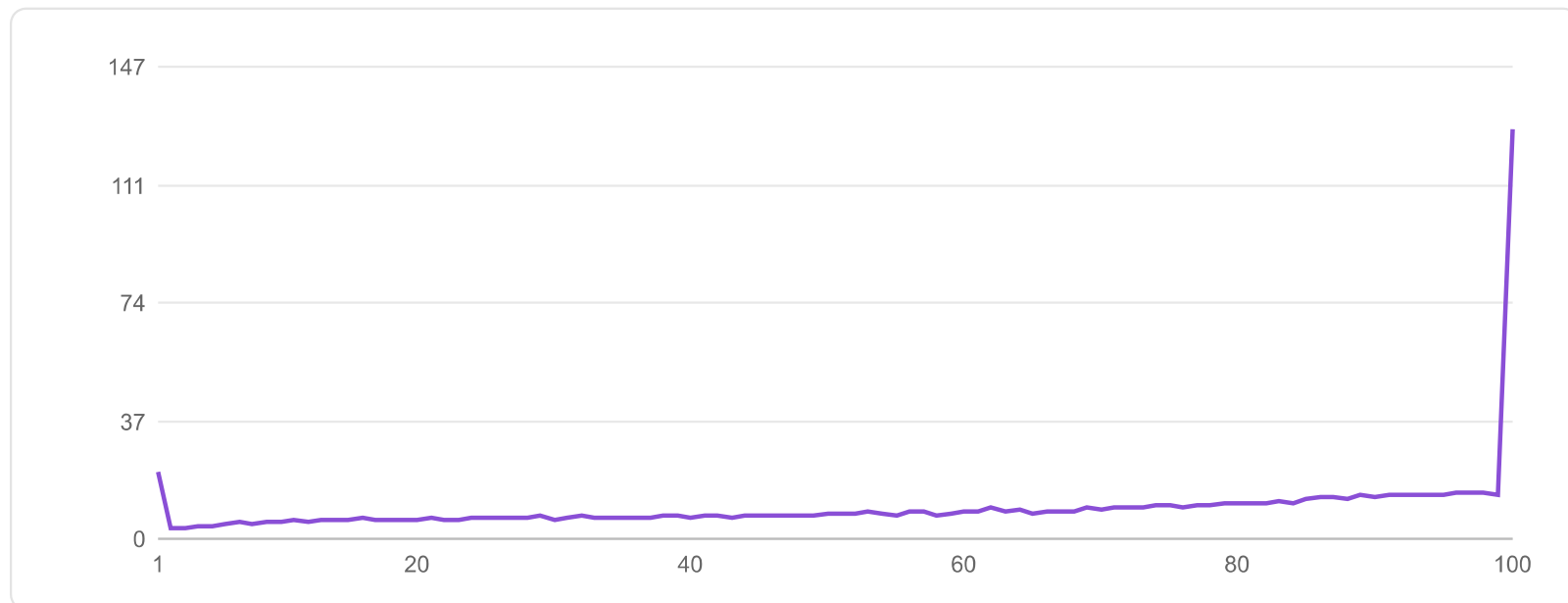
Analogous to the STAR/HISAT2 alignment bar in an nf-core/rnaseq MultiQC report: uniquely mapped, multi-mapped and resolved by EM, mapped but not overlapping an annotated gene, and unmapped.

Per-base mean quality (raw, pre-trim)



X axis: position in read (1-based). Y axis: mean Phred quality. Analogous to FastQC's "per base sequence quality" plot.

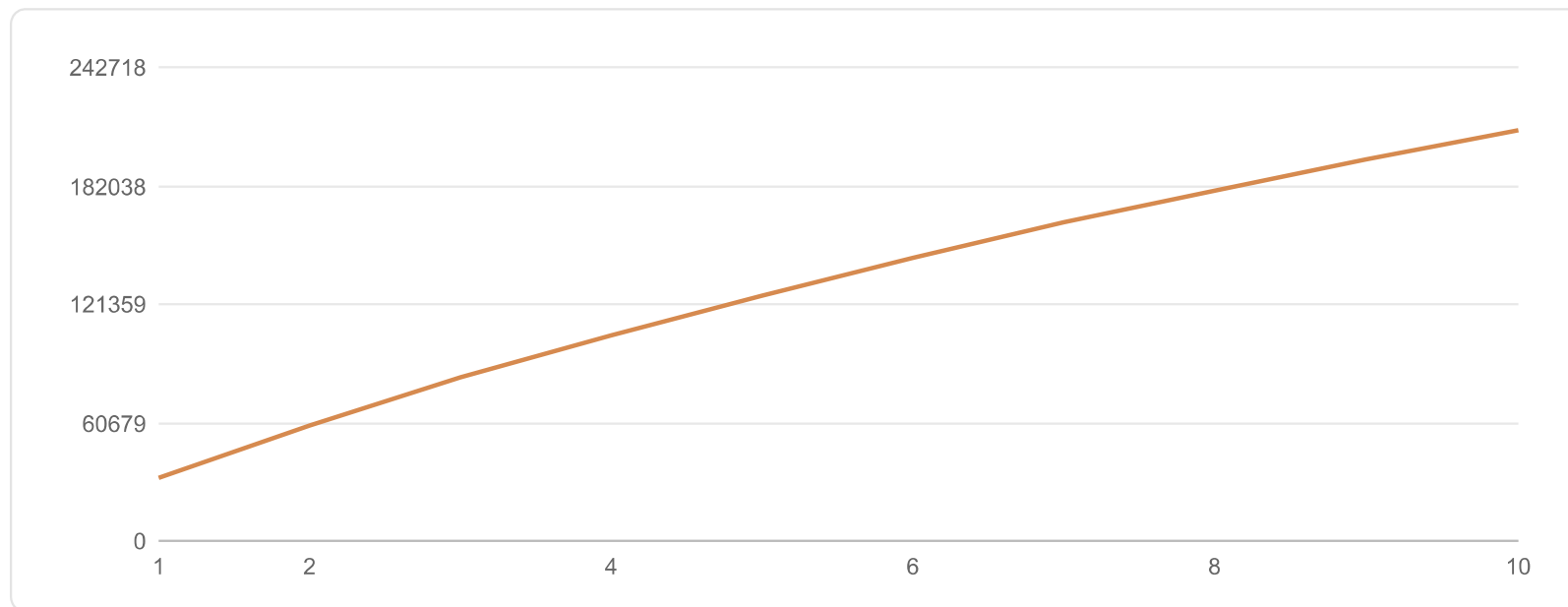
Gene-body coverage



X axis: position along gene body, 0% = 5' end, 100% = 3' end (strand-corrected). Y axis: relative read density, averaged across 3996 genes with ≥ 10 uniquely-assigned reads each, each gene's own curve normalized to sum to 1 before averaging so highly-expressed genes don't dominate the shape.

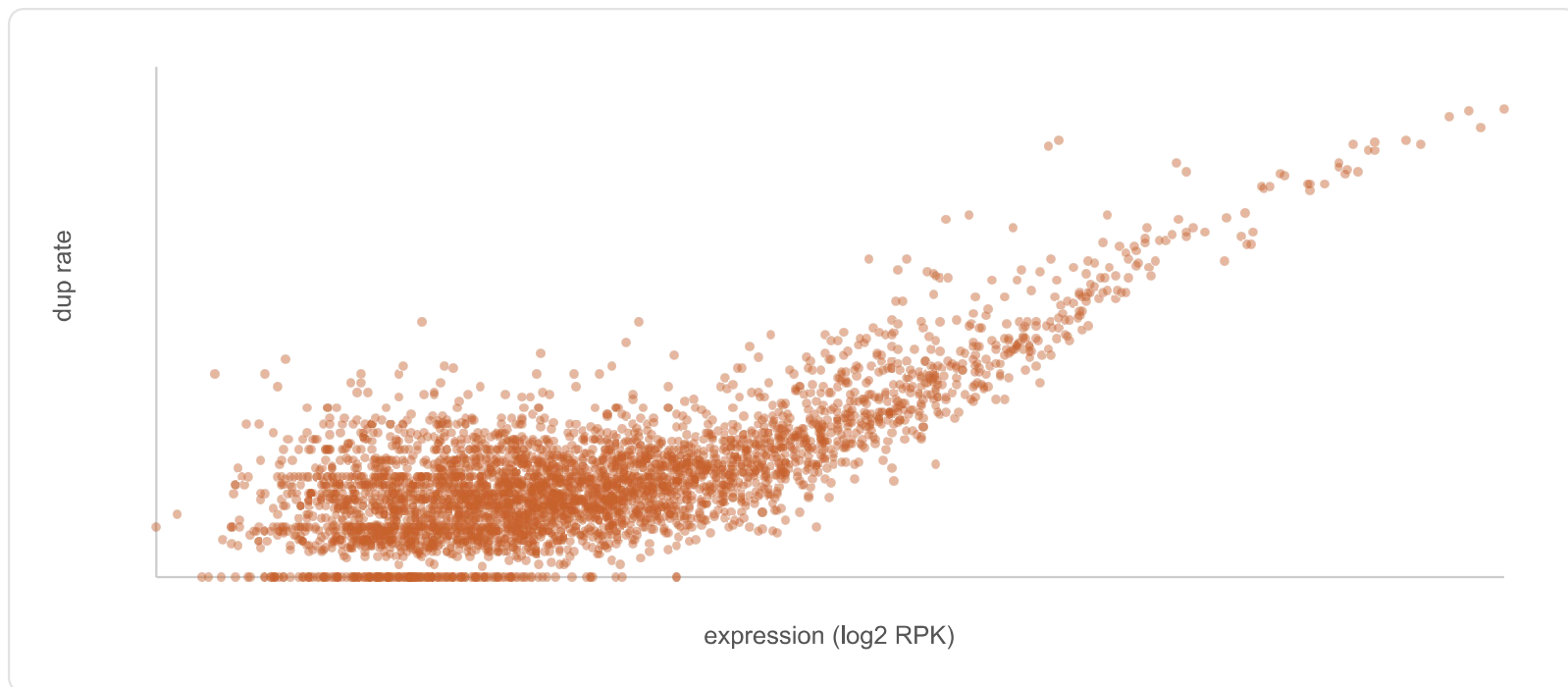
Analogous to RSeQC's geneBody_coverage.py plot -- a flat curve indicates uniform coverage; a skew toward either end can indicate RNA degradation (3' skew) or other library-prep biases.

Library complexity (saturation curve)



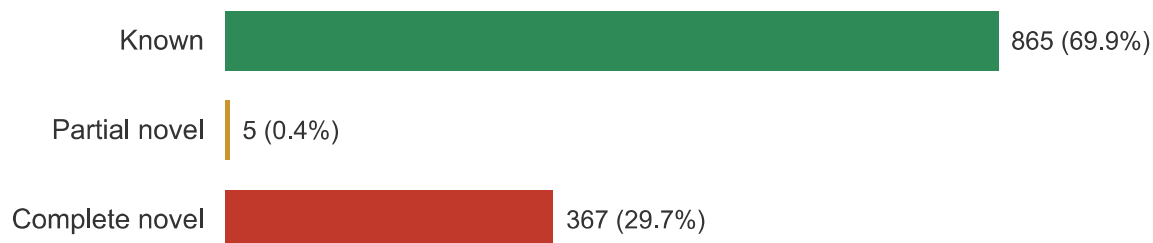
X axis: subsample depth, 10% to 100% of all mapped fragments. Y axis: distinct (chromosome, position, strand) combinations found at that depth -- i.e. how many unique molecules this many reads would have captured, computed on a fixed random shuffle of the real mapped reads (not simulated). Still rising steeply at 100% means deeper sequencing would likely find substantial new material; a flattening curve means this library is close to saturated at the depth sequenced here. Analogous to Preseq's c_curve, though this describes the observed depth's shape rather than Preseq's own statistical extrapolation beyond it.

Duplication vs. expression (dupRadar-equivalent)



Each point is one gene (≥ 10 uniquely-assigned fragments, position-based duplicates: identical chrom+position+strand, Picard-style -- computed independently of --umi-len). Pearson $r = 0.738$. A rising trend (duplication increasing with expression) is the expected pattern for ordinary PCR-driven duplication; genes far above the trend for their expression level are candidates for a library-prep artifact rather than real signal. Analogous to dupRadar's duplication-rate-vs-expression plot. Full per-gene values: dupradar.tsv.

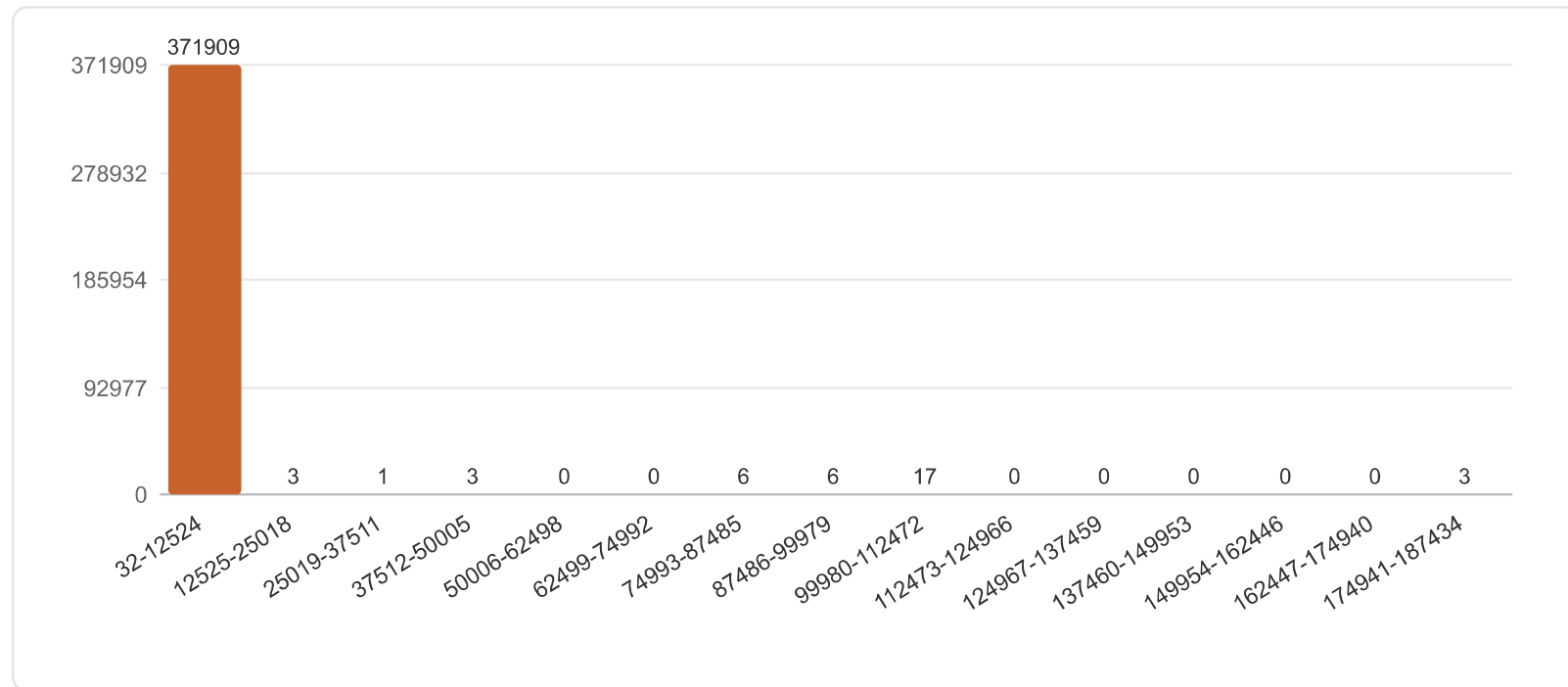
Junction annotation (RSeQC-equivalent)



1237 splicing event(s) (one per spliced alignment) classified against 364 splice junction(s) annotated in the GTF's exon rows -- known: both splice sites match an annotated junction as a pair; partial novel: one site is annotated but paired with an unannotated site; complete novel: neither site is

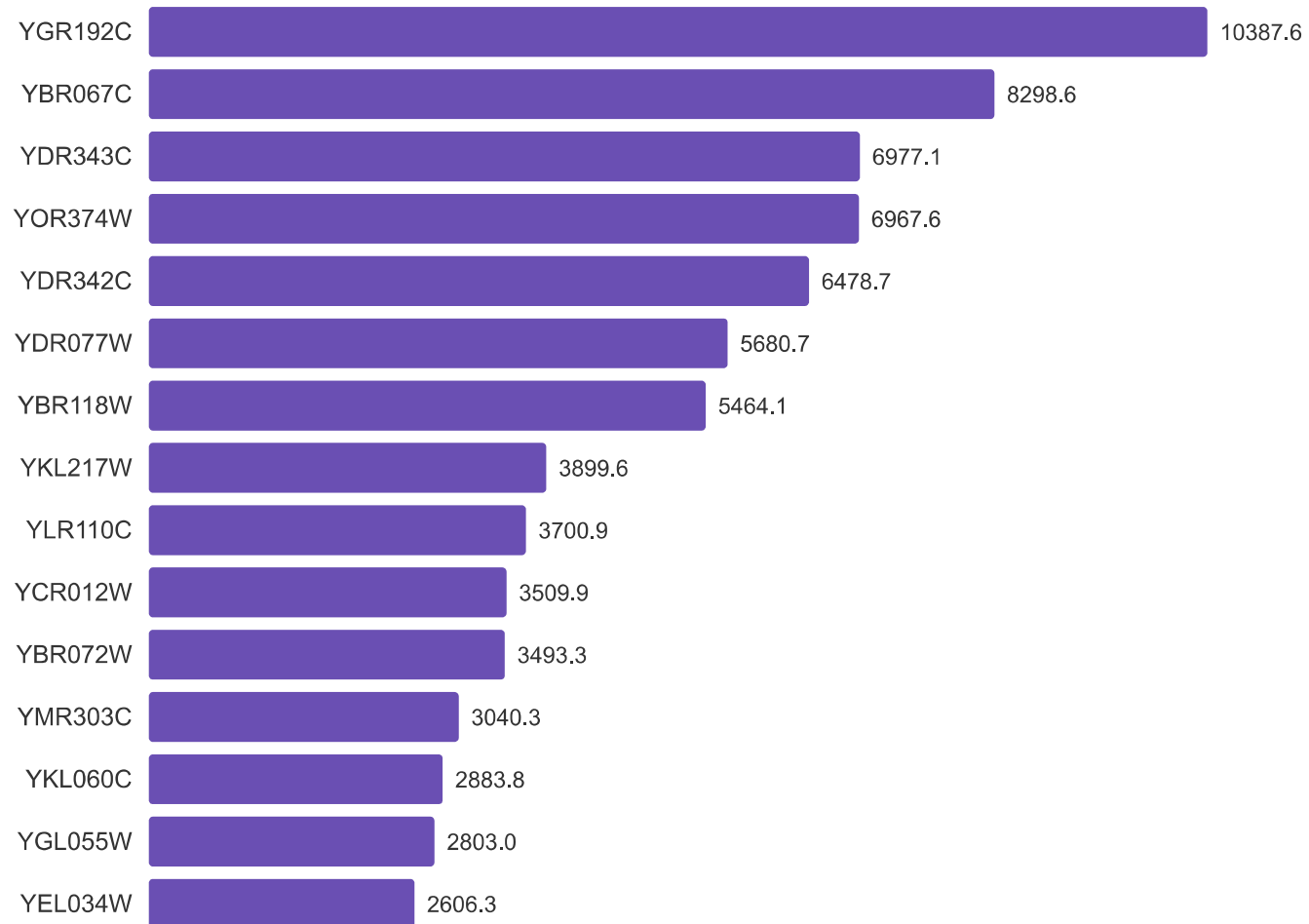
annotated. Same three-way classification RSeQC's junction_annotation.py reports. Full per-junction list (chrom, intron start/end, category, supporting-read count): junctions.tsv.

Fragment size distribution



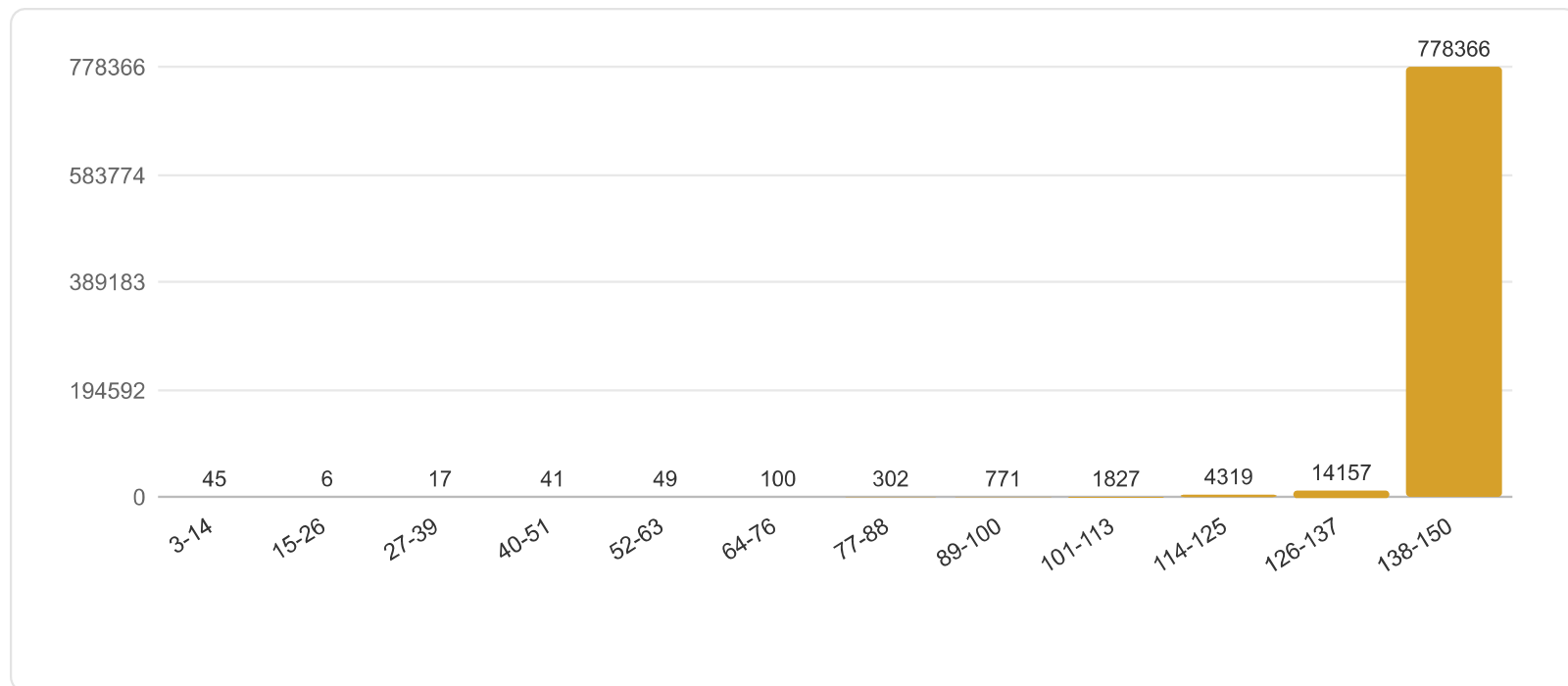
Genomic span of properly-paired fragments (FR orientation, same chromosome). Analogous to RSeQC's "inner distance" plot in the nf-core/rnaseq MultiQC report.

Top 15 expressed genes



Ranked by EM effective count (unique reads + proportional multi-mapper share), same metric as gene_counts.tsv.

Trimmed read length distribution



Length after adapter/quality trimming. Analogous to TrimGalore's length-distribution plot in the MultiQC report.

Not included here

nf-core/rnaseq's MultiQC report also includes a DESeq2 PCA plot and a sample-distance heatmap. Both compare gene expression *across multiple samples* (they need ≥ 2 samples to place points or compute pairwise distances), so they don't apply to a single run of this pipeline any more than they'd apply to a single sample run through nf-core/rnaseq — they belong in whatever aggregates multiple runs' gene_counts.tsv files, not in a per-run report like this one.